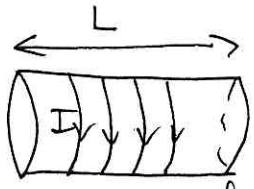
Self-inductance and LR circuit



turns per unit length

$\vec{B} = \mu_0 n I$

for 1 loop: $\Phi_1 = B\pi R^2 = \mu_0 n I \pi R^2$

for $n \cdot L$ loops: $\Phi = \mu_0 n^2 L \pi R^2 I$

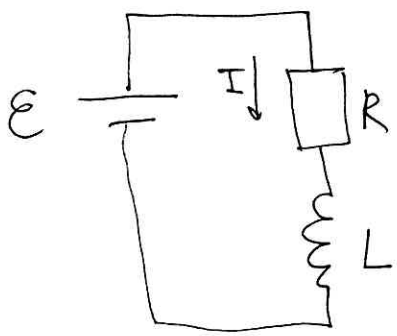
$\Rightarrow$ self-inductance $L = \frac{\Phi}{I} = \mu_0 \pi R^2 n^2 L$

determined by geometric quantities only!

Suppose $I = I(t) \Rightarrow \Phi = \Phi(t) \Rightarrow \mathcal{E}_{back} = -\frac{d\Phi}{dt}$ is generated.

(back emf)

$\mathcal{E}_{back} = -L\frac{dI}{dt}$



$$\mathcal{E} - L\frac{dI}{dt} = IR$$

$$\frac{dI}{dt} + \frac{R}{L}I = \frac{\mathcal{E}}{L}$$

$$\Rightarrow I(t) = \frac{\mathcal{E}}{R} + k e^{-Rt/L}$$

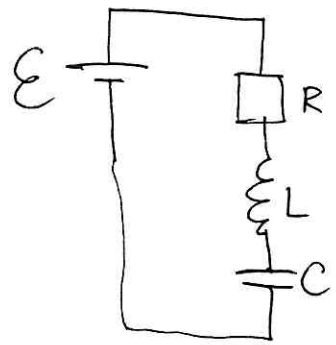
indeed $-k\frac{R}{L}e^{-Rt/L} + \frac{R}{L}\left(\frac{\mathcal{E}}{R} + k e^{-Rt/L}\right) = \frac{\mathcal{E}}{L}$

Initial condition: $I(t=0) = 0 \Rightarrow k = -\frac{\mathcal{E}}{R}$

$$I(t) = \frac{\mathcal{E}}{R}\left(1 - e^{-tR/L}\right)$$

$\tau = \frac{L}{R}$ = time constant of a circuit.

AC circuit



$V = IR$

$V = L\frac{dI}{dt}$

$V = \frac{q}{c} = \frac{1}{C}\int_0^t dt\, I(t)$ (assuming $I=0$ for $t<0$)

$$\mathcal{E} = IR + L\frac{dI}{dt} + \frac{1}{c}\int_0^t I\,dt = IR + L\frac{dI}{dt} + \frac{q(t)}{C}$$

Suppose $I(t) = I_0 e^{i\omega t}$, $\mathcal{E}(t) = V_0 e^{i\omega t}$ } we actually need only <u>real</u> part of these for physical quantities.

$$\int e^{i\omega t}dt = \frac{e^{i\omega t}}{i\omega}$$

$$I(t) = \frac{dq(t)}{dt} \Rightarrow q(t) = \frac{I_0 e^{i\omega t}}{i\omega}$$

$$V_0 e^{i\omega t} = \mathcal{E}(t) = I_0 R e^{i\omega t} + L i\omega I_0 e^{i\omega t} + \frac{I_0 e^{i\omega t}}{i\omega C}$$

$$V_0 = I_0 \underbrace{\left(R + i\omega L + \frac{1}{i\omega C}\right)}_{Z}$$

Z = impedance (complex!)

generally, there is <u>phase shift</u> between $\mathcal{E}(t)$ and $I(t)$.

$$I_0 = \frac{V_0}{R + i\omega L + \frac{1}{i\omega C}}$$

The amplitude of the current $|I_0| = \dfrac{|V_0|}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$

$\Rightarrow$ there is a <u>resonance frequency</u> $\omega_0 = \frac{1}{\sqrt{LC}}$

when $|I_0| = \frac{|V_0|}{R}$ has maximum.

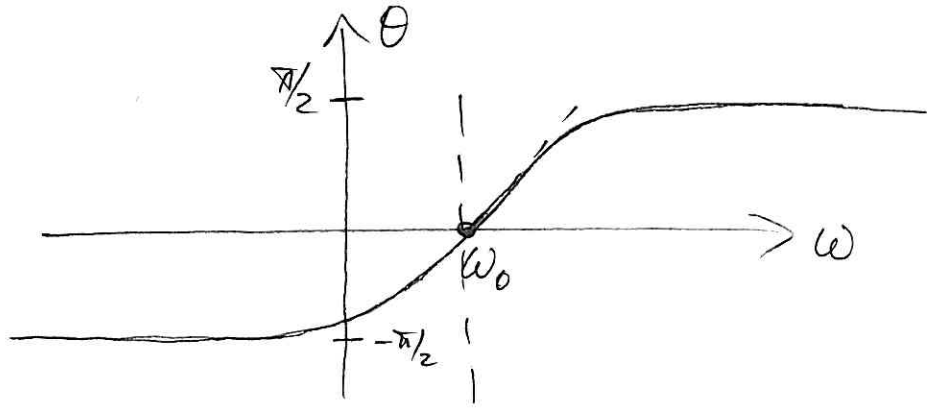
Also, $V_0 = I_0 |Z| e^{i\theta}$, θ = phase-lag.

$$|Z| e^{i\theta} = |Z|(\cos\theta + i\sin\theta) = R + i\left(\omega L - \frac{1}{\omega C}\right)$$

$$\left.\begin{aligned} \Rightarrow R &= |Z|\cos\theta \\ \omega L - \frac{1}{\omega C} &= |Z|\sin\theta \end{aligned}\right\} \tan\theta = \frac{\omega L - \frac{1}{\omega C}}{R}$$

$$\theta = \tan^{-1}\left(\frac{\omega L - \frac{1}{\omega C}}{R}\right) = \tan^{-1}\left(\frac{\frac{\omega^2}{\omega_0^2} - 1}{\omega C R}\right)$$

$\Rightarrow$



$$\omega L - \frac{1}{\omega C} = \frac{1}{\omega C}\left(\omega^2 L C - 1\right) = \frac{\omega^2/\omega_0^2 - 1}{\omega C}$$

<u>Dissipated power</u> $V(t) = V_0 e^{i\omega t}$

$$I(t) = I_0 e^{i\omega t - i\theta}$$

$$\Rightarrow V(t) = I(t) Z = I(t) |Z| e^{i\theta}$$

$P = \langle V(t)\, I(t) \rangle$ where the average is over the period of oscillation

$$P = \frac{1}{(2\pi/\omega)} \int_0^{2\pi/\omega} V(t)\, I(t)\, dt = V_0 I_0 \frac{\omega}{2\pi} \int \cos(\omega t) \cos(\omega t - \theta)\, dt$$

$(\omega t = x)$

$$= \frac{\omega}{2\pi} \cdot \frac{V_0 I_0}{\omega} \int_0^{2\pi} \cos x \cos(x-\theta)\, dx =$$

$$= \frac{V_0 I_0}{2\pi} \int_0^{2\pi} \cos x \left(\cos x \cos\theta + \sin x \sin\theta \right) dx =$$

$$= \frac{V_0 I_0}{2\pi} \left\{ \underbrace{\left(\int_0^{2\pi} \cos^2 x\, dx \right)}_{\pi} \cos\theta + \sin\theta \underbrace{\int_0^{2\pi} \cos x \sin x\, dx}_{\phi} \right\}$$

$$= \frac{1}{2} V_0 I_0 \cos\theta = \frac{1}{2} V_0 I_0 \frac{R}{|Z|} = \frac{1}{2} I_0^2 R$$

$$\cos\theta = \frac{R}{|Z|}$$

$$P = \frac{1}{2} V_0 I_0 \frac{Re(Z)}{|Z|} = \frac{1}{2} I_0^2\, Re(Z)$$

$$= \frac{1}{2} Re(Z) \frac{V_0^2}{|Z|^2}.$$

$$\frac{V_0}{|Z|} = I_0$$

The power is determined only by <u>real</u> part of Z ; $Re(Z) = R$.

$\Rightarrow$ (ideal) inductor and capacitor does not dissipate <u>power</u>!